University of Asia Pacific

Admit Card

Mid-Term Examination of Fall, 2025

Financial Clearance	PAID

Registration No : 22201243

Student Name : H. M. Tahsin Sheikh

Program : Bachelor of Science in Computer Science and Engineering

SI.NO.	COURSE CODE	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
1	CSE 400-A	Final Year Design Project	3.00	
2	CSE 401	Operating System	3.00	
3	CSE 402	Operating System Lab	1.50	
4	CSE 405	Numerical and Mathematical Analysis for Engineer	3.00	
5	IMG 401	Industry and Operational Management	3.00	
6	CSE 413	Machine Learning and Deep Learning	3.00	
7	CSE 437	Robotics	3.00	
8	CSE 414	Machine Learning and Deep Learning Lab	1.50	
9	CSE 438	Robotics Lab	1.50	

Total Credit: 22.50

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.

2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.

3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.

4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall.
Violators will be subjects to disciplinary action.

This is a system generated Admit Card. No signature is required.

Admit Card Generation Time: 24-Mar-2026 09:03 PM

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 25
Program: B.Sc. in CSE
4^{th} year 1^{st} Semester

Course Title: Operating Systems — Course Code: CSE 401
Time: 1 hour. — Credit Hour: 3:00 — Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. We are running various processes on our computers all the time. How does the operating system handle this situation to prevent overload or crash? [3] CLO1/PLOa

b. Draw the Gantt chart and find the average waiting time using the appropriate algorithm for the processes shown in **Table 1**. (Quantam=3) [7] CLO2/PLOb

Table 1. Process Information

Process	Burst Time
P1	13
P2	1
P3	9
P4	14
P5	19

QUESTION 2 [10 MARKS]

a. Imagine a web server application is running on a Linux system. When a new client connects, the server creates a duplicate of itself so the new process can handle the client request while the original server continues listening for other clients. Explain what happens inside the operating system when this duplicate process is created using the `fork()` system call. [3] CLO1/PLOa

b. Compare different multi-threading models and discuss which model is more efficient or preferable in operating systems. [3] CLO1/PLOa

c. Consider a system in which multiple producers place items into a bounded buffer and a consumer removes items from it. If the buffer is full, producers must wait; if it is empty, the consumer must wait. Apply semaphores to solve this Producer–Consumer problem. [4] CLO2/PLOb

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 25
Program: B.Sc. in CSE
4^{th} year 1^{st} Semester

Course Title: Machine Learning and Deep Learning | Course Code: CSE 413
Time: 1 hour. | Credit Hour: 3.00 | Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [8 MARKS]

a. A data scientist team is trying to build a suitable machine learning model to predict house prices using features such as house size, number of bedrooms, and location. They experiment with two algorithms, that produce two different models (A and B): [4] CLO1/PLO1

- Model A makes very simple assumptions about the data and consistently misses important patterns and under fits the data.
- Model B is very complex and performs extremely well on training data but performs poorly on new unseen data.

Explain how the concepts of **bias** and **variance** relate to these two models.

b. Suppose a machine learning engineer is training a model where the loss function to minimize is: [4] CLO2/PLO3

$$f(x)=2x^2 + 3x + 1$$

To find the optimal parameter value, the engineer decides to use Gradient Descent with:

- Initial parameter value $x = 0$
- Learning rate $\alpha = 0.25$

Using the Gradient Descent update rule, calculate the value of x after three iterations. Show all the intermediate steps and calculations, including the gradient of the function and the updated value of x at each iteration.

QUESTION 2 [12 MARKS]

a. An online electronics retailer wants to predict whether a customer will buy a laptop based on three features: Age, Income, and Student status. The marketing team collected a small historical dataset shown in **Table 1**. The company now wants to classify a new customer with the following test data: [5] CLO2/PLO2

Test Data: Age = Young, Income = High, Student = Yes

Using the Naïve Bayes classification method, determine whether the new customer will BuyLaptop (Yes/No). Calculate the frequency table, likelihood table and show all the detailed calculation steps.

Table 1: Customer Records

Age	Income	Student	BuyLaptop
Young	High	No	No
Young	High	No	No
Young	Low	Yes	Yes
Old	High	No	Yes
Old	Low	Yes	Yes
Old	High	Yes	Yes
Young	Low	Yes	No

b. A healthcare company has developed a machine learning model to classify patients into two health categories based on medical test results: Fit, and Unfit. Before deploying the system in hospitals, the team wants to evaluate the performance of the classification model. After testing the model on a validation dataset, the confusion matrix shown in **Figure 1** was obtained. [5+ 2] CLO2/PLO2

		Predicted Class	
	Total = 8+4 = 12	**Fit** 7	**Unfit** 5
Actual Class	**Fit** 8	TP 6	FN 2
	Unfit 4	FP 1	TN 3

Figure 1: Confusion Matrix

i) Using the above confusion matrix, **evaluate the performance** of the model by calculating the metrics: Accuracy, Precision, Recall (Sensitivity), Specificity and F1-score.

ii) Briefly **interpret** the results to determine whether the model performs well for this classification task.

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 25
Program: B.Sc. in CSE
4th year 1st Semester

Course Title: Robotics | Course Code: CSE 437
Time: 1 hour. | Credit Hour: 3.00 | Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

A 500-bed multi-specialty hospital faces frequent delays in medicine delivery due to nurse shortages and heavy patient load. During peak hours, nurses spend nearly 30–40% of their time transporting medicines, lab samples, and documents between pharmacy, ICU, and wards. To reduce workload and minimize infection transmission risk, the hospital needs a robot operating autonomously across hospital floors.

a. Describe the target and functionalities of the robot you plan to design. Also explain the input and output peripherals you will use and justify why you selected them. [6] CLO1/PLO1

b. Draw the state machine diagram representing the operation of your robot and explain the diagram. [4] CLO2/PLO5

QUESTION 2 [10 MARKS]

A large rice field in a semi-rural region suffers from inconsistent irrigation and late disease detection, leading to 20% yield loss annually. The farmer needs a robot that moves across predefined paths to monitor soil condition, crop health, automate spraying, pluck weeds and collect harvest.

a. Describe the objective and tasks of your robot system and illustrate the corresponding embedded system diagram. [6] CLO1/PLO3

b. Develop the pseudocode that describes how your robot operates and provide an explanation [4] CLO3/PLO3

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 25
Program: B.Sc. in CSE
4^{th} year 1^{st} Semester

Course Title: Numerical and Mathematical Analysis for Engineers Course Code: CSE 405
Time: 1 hour. Credit Hour: 3.00 Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. Describe the sources of errors in numeral methods. What problems can be created by round off errors in mathematical solutions? [2] CLO1/PLO2

b. Explain how the bisection method guarantees at least one root for a continuous function in an interval $[a, b]$. [2] CLO2/PLO2

c. Demonstrate the root of the given equation $x^2\text{-}3x = 3$ for $x \in [1, 3]$ using the bisection method. Conduct three iterations to estimate the root of the above equation. Also find the absolute relative approximate error at the end of each iteration. [6] CLO3/PLO1

QUESTION 2 [10 MARKS]

a. Central, forward, and backward finite differences are three common methods for approximating the first derivative of a smooth function at a point. Explain which of these methods is generally more accurate and more rapidly convergent as the step size h decreases, and why? [3] CLO3/PLO1

b. The velocity of a rocket is given by [7] CLO3/PLO1

$$v(t) = 2500 \ln\left[\frac{18 \times 10^4}{18 \times 10^4 - 2600t}\right] - 9.5t, \quad 0 \leq t \leq 30$$

Calculate the jerk at t=18 s. Use a step size of Δt=2s. Use the central difference approximation of the second derivative.

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination, Fall 25
Program: B.Sc. in CSE
4th Year 1st Semester

Course Title: Industry and Operational Management | Course Code: IMG 401
Time: 1 hour. | Credit Hour: 3.00 | Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. List the key differences between software operations and hardware operations in terms of flexibility, cost structure, and iteration speed. [3] CLO1/ PLO a

b. A smartphone manufacturing company releases a new model, but many devices fail within a few days due to defective screens. Later investigation reveals that several quality inspection steps were skipped during production to speed up delivery. Explain the core element of operations management neglected in this scenario. [3] CLO1/ PLO a

c. A newly promoted supervisor in a manufacturing company has strong technical knowledge about machines, but team members are demotivated, communication is poor, and the supervisor struggles to understand how different departments work together to achieve organizational goals. Discuss the fundamental management skills the supervisor needs to improve in this situation and their impact. [4] CLO1/ PLO a

QUESTION 2 [10 MARKS]

a. Explain the democratic leadership style by discussing its key advantages, disadvantages, and situations where it is most suitable. [5] CLO2/ PLO i

b. Two departments in an organization are in conflict over limited project resources. One manager suggests that both sides give up some demands to reach an agreement quickly, while another suggests working together to find a solution that satisfies both sides. Analyze which of the two conflict management strategies would be more effective in your opinion. [5] CLO2/ PLO i